

CONTACT	Email: vivwhite@cs.ubc.ca Website: vivianwhite.github.io LinkedIn: linkedin.com/in/vivian-white
RESEARCH INTERESTS	I am passionate about furthering our scientific understanding of black-box deep networks to develop robust, interpretable, and efficient models for computer vision. I currently study (1) interpretability and (2) energy efficiency of deep models during test-time adaptation.
PUBLICATIONS	3. V White, M Chaudhary, G Wolf, G Lajoie, KD Harris. Learning Stochastic Rainbow Networks. NeurIPS Workshop on Scientific Methods for Understanding Deep Learning (Sci4DL), 2024. 2. V White, A White, J Wild, T Nguyen, F Huang. Human Error Scenario Analysis of Software Defects. ISSRE Workshop on Human Factors for Software Dependability, 2024. 1. V White, M Chaudhary, G Wolf, G Lajoie, KD Harris. Learning and Aligning Structured Random Feature Networks. ICLR Workshop on Representational Alignment, 2024.
EDUCATION	The University of British Columbia (UBC), Vancouver, British Columbia, CA. · Ph.D., Computer Science, expected 2030 · Certificate in Instructional Skills for Graduate Students Western Washington University (WWU), Bellingham, Washington, USA. · M.S., Computer Science, June 2025, GPA 4.00/4.00 · B.S., Computer Science, June 2024, GPA 3.78/4.00 · Minors in Mathematics and Honors Interdisciplinary Studies
GRANTS AND AWARDS	Graduate · Maria Klawe Service Award, UBC CS Dept., 2026 · AML-TN Funded Trainee, UBC CS Dept., 2025 · CS Merit Scholarship, UBC CS Dept., 2025 · Outstanding MS Graduate in CS, WWU, 2025 · Dean J. Alan Ross Travel Fund Award, WWU, 2024 · CS Graduate Fellowship, WWU CS Dept., 2024 · Graduate Recruitment Tuition Waiver, WWU, 2024 Undergraduate · Outstanding BS Graduate in CS, WWU, 2024 · 1st-place for CS poster presentation, Emerging Researchers National (ERN) Conference, 2024 · Barbara Ellen Maguire-Veith Family Scholarship, WWU CS Dept., 2023 · Fellow, International Network for Bio-Inspired Computing (IN-BIC), 2023 · Lars and Elaine Giusti Scholarship for CS, WWU CS Dept., 2022 · CS/Math Distinguished Scholar Award, WWU CS Dept., 2020-2023 · Western Foundation Distinguished Scholar Award, WWU, 2020 · Merit Scholarship, WWU, 2020 · Admissions Achievement Award, WWU, 2020 · Admissions Annual Scholarship, WWU, 2020 · Scholarship Award, WSECU, 2020
EXPERIENCE	Graduate Research Assistant, UBC and Vector Institute for AI, Sept 2025–present · Studying interpretability and efficiency of test-time adaptation with Dr. Evan Shelhamer · Supported by UBC’s Advanced Machine Learning – Training Network (AML-TN) trainee award Masters Intern, Pacific Northwest National Laboratory (PNNL), June–Aug 2025 · Worked on Math for AI Assurance with Dr. Henry Kvinge · Fine-tuned a base LLM using Group Relative Policy Optimization for statistical weight analysis Graduate Teaching Assistant, WWU, Dec 2024–June 2025 · Taught weekly labs and graded for an introductory data science class · Graded for a graduate analysis of algorithms class Graduate Research Assistant, WWU, June 2024–June 2025

- Developed stochastic rainbow networks with Drs. Kameron Harris, Guy Wolf, and Guillaume Lajoie

Undergraduate Research Assistant, WWU, June 2022–June 2024

- Developed randomized scattering convolutional networks and learnable structured random features with Drs. Kameron Harris, Guy Wolf, and Guillaume Lajoie

Research Intern, Mila - the Quebec AI Institute, Montréal, QC, July–Sept 2023

- Co-advised by Drs. Guy Wolf and Guillaume Lajoie studying learnable structured random features
- Funded by IN-BIC fellowship grant through National Science Foundation (NSF) AccelNet program

Undergraduate Research Assistant, WWU, April 2021–March 2022

- Worked with Dr. Scott Wehrwein on project applying machine learning techniques to international border legibility tasks for computer vision

RESEARCH TALKS *What Changed? Interpreting Model Updates During Training and Test-Time Adaptation*
 · Canadian Celebration of Women in Computing, Vancouver, BC, Nov 2025

Analysis of LLM Weights During RL Reasoning Finetuning
 · Gold Intern Symposium, PNNL, virtual, Aug 2025

Learning Stochastic Rainbow Networks
 · Graduate Research Symposium, WWU, Bellingham, WA, June 2025
 · ERN Conference, Atlanta, GA, Mar 2025

Learning and Aligning Structured Random Feature Networks
 · NeuroAI Workshop, University of Oregon, Eugene, OR, Aug 2024

Unveiling the Cognitive Roots: Human Error Scenario Analysis of Software Defects
 · Senior Project Symposium, WWU, Bellingham, WA, May 2024

POSTER
 PRESENTATIONS *Green TTA: Benchmarking the Energy Efficiency of Test-Time Adaptation*
 · AI/CRV, Burnaby, BC, May 2026

What Changed? Interpreting Model Updates During Training and Test-Time Adaptation
 · ACM Celebration of Cascadia Women in Computing, Vancouver, BC, Oct 2025

Learning Stochastic Rainbow Networks
 · NeurIPS Workshop on the Science of Deep Learning, Vancouver, Canada, Dec 2024

Learning and Aligning Structured Random Feature Networks
 · CoNectome Symposium, University of Washington, Seattle, WA, May 2024
 · ICLR Workshop on Representational Alignment, Vienna, Austria, May 2024
 · ERN Conference, Washington DC, Mar 2024

Randomized Scattering Convolutional Networks
 · DeepMath Conference, Johns Hopkins University, Baltimore, MD, Nov 2023
 · NeuroAI Workshop, Mila, Montréal, QC, Oct 2023
 · NSF S-STEM Scholars Meeting, Washington DC, Sept 2023
 · NeuroAI Workshop, University of Washington, Seattle, WA, Sept 2022

SERVICE · Reviewer: NeurIPS 2025, Sci4DL Workshop @ ICLR 2026 (Outstanding Reviewer), TTU Workshop @ ICLR 2026
 · VP of Academics, UBC CSGSA, 2026-2027
 · Co-Organizer, AI/CRV Workshop on AI for Sustainability and Science, 2026
 · Co-Founder, UBC Graduate Women in Computer Science (GWICS), 2025-

TECHNICAL SKILLS Languages: Python (PyTorch, HuggingFace, numpy, pandas, scikit-learn), Java, C, Javascript
 Systems: Bash, Linux, L^AT_EX, git, HPC cluster environments, AWS

SOFT SKILLS Event planning, leadership, communication, community building

REFERENCES

Evan Shelhamer

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CIFAR AI Chair, Vector Institute for AI
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Kameron Decker Harris

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